Disentangling the Cosmic Web: Characterizing Relationships among Cosmic Structures [PHYS073]

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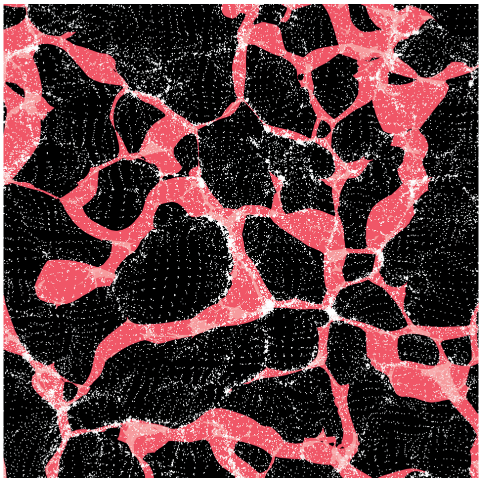
Research Question:

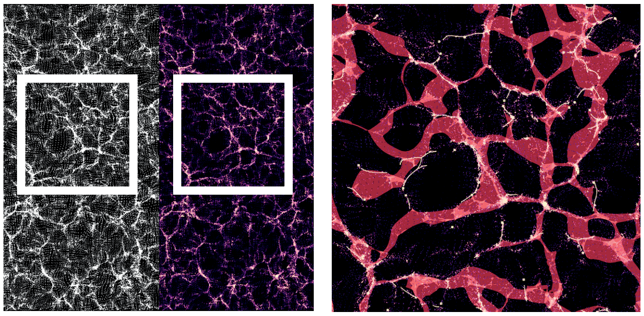
How do walls fit into the cosmic web? How does their density and spatial distribution compare to other structures?



Walls (red) overlaid on an N-body simulation in 200 x 200 x 10 Mpc3 (16 million light years thick) slab flattened into 2D.
Image created by Emma Fuleky using Paraview 2026

Methodology:

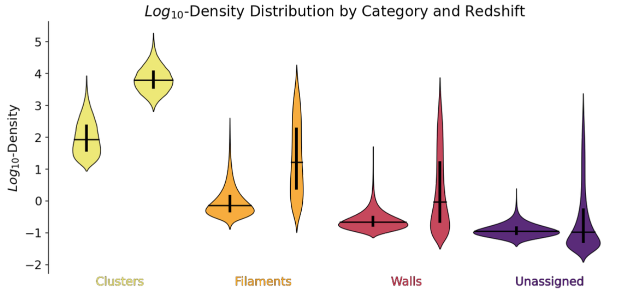
Sim→Crop→Density→Persistence→Structure→Statistics



Left: Original particle simulation in a 10 Mpc3 thick slab (left half) and the same slab colored by log-density (right half). Right (enlarged area of the white box, sized 200 x 200 x 10 Mpc3): Walls (red), filaments (orange), and clusters (yellow).
Image created by Emma Fuleky using Paraview 2026

Results:

- Over time each structure got denser
- Density of structures is inversely related to their dimensionality



	Clusters z=3	Clusters z=0	Filaments z=3	Filaments z=0	Walls z=3	Walls z=0	Unassigned z=3	Unassigned z=0
max	3.937	5.276	2.609	4.275	1.714	3.878	0.393	3.378
q75	2.398	4.093	0.195	2.301	−0.472	1.246	−0.799	−0.233
mean	2.004	3.822	−0.012	1.347	−0.608	0.333	−0.918	−0.584
median	1.935	3.800	−0.132	1.222	−0.665	−0.026	−0.943	−0.975
q25	1.551	3.524	−0.352	0.359	−0.816	−0.686	−1.067	−1.322
min	0.944	2.814	−0.887	−0.985	−1.150	−1.503	−1.425	−1.917
sd	0.562	0.408	0.496	1.183	0.310	1.241	0.215	1.067

Each violin-plot shows the distribution of log10-density for a cosmic structure at z=3 (left) and z=0 (right). Below each plot are the corresponding summary statistics.
Graph and table created by Emma Fuleky using python 2026

Conclusions:

- While filaments have higher density than walls, walls contain more particles than filaments
- Results challenge the literature's neglect of walls and prove that walls are structurally fundamental in the cosmic web.

Abstract:

Gravity and expansion shape the universe into a vast cosmic web of empty regions (voids), flat sheets (walls), thin threads (filaments), and dense clusters. While filaments and clusters are well studied, walls are harder to detect and therefore less understood. Because walls and filaments form at different stages of gravitational collapse, comparing their density and spatial relationship can improve our understanding of how dark matter organizes large-scale cosmic structures. <u>My project automates the full workflow of extracting, analyzing, and visualizing the cosmic web (clusters, filaments, walls) from cosmological snapshots</u>. The pipeline splits the data into tiled volumes of manageable size, executes the components of DisPerSE (Delaunay Tessellation Field Estimator and Morse–Smale Topology Extractor), produces per-crop statistics, 2D/3D visualizations, and aggregates results across the tiles. Results show that these structures have become more pronounced over time due to gravitational collapse. As expected, the share of particles associated with each of these structures has increased between redshifts z=3 and z=0, while the number of particles not assigned to any of these categories (voids) has declined. Additionally, the log-density distribution in categories other than clusters has widened, mostly by extending towards higher-density values. These findings allow us to understand the material skeleton of the universe and the gravitational backbone shaped by dark matter. My future research will extend this analysis of to simulations from different cosmological models.

Project website:

- Project details
- Workflow, including a video with 3D visualizations
- User guides
- Results, including poster, video summary

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